1 Personal information

Christoph Benzmüller

Full Professor
U Bamberg
Chair for AI Systems Engineering

Professor (apl.)
FU Berlin
Dep. of Maths and Computer Science

Contact address:
An der Weberei 5
96047 Bamberg
Germany

christoph.benzmueller@uni-bamberg.de
`http://christoph-benzmueller.de/`

Luca Pasetto

Postdoctoral researcher
University of Luxembourg
Department of Computer Science, Computational Law and Machine Ethics group

Contact address:
Maison du Nombre
6, avenue de la Fonte
L-4364 ESCH
Luxembourg

luca.pasetto@uni.lu
`https://lucapasetto.github.io/`

2 General proposal information

Title

Experimenting with the LogiKEy Framework & Methodology: Normative Reasoning, Computational Metaphysics, and More

Category

Logic and Computation
Introductory, could also be advanced if that would better fit the program.

3 Contents information

3.1 Abstract

The course provides an introduction to the LogiKEy framework and methodology for Logic and Knowledge Engineering. LogiKEy's unifying approach is based on semantical embeddings of object logics in expressive classical higher-order logic (HOL). We begin with the methodology, introduce HOL, and give a guided practical tour of the Isabelle/HOL proof assistant. Next, we explain the technique of semantical embeddings of object logics in the meta-logic HOL, which enables the automation and model finding available in HOL provers to reason with the embedded logics. We then present two applications of the framework: (1) encoding deontic logics and corresponding ethical or legal theories to support computational tools for normative reasoning; and (2) the analysis of a version of the ontological argument from Kurt Gödel's papers. We conclude with topics in knowledge representation within LogiKEy: combining logics, and encoding agents,

actions, obligations, and knowledge. As a case study, we discuss the formalization of epistemic rights and present the embedding of a dynamic logic of the right to know in HOL.

3.2 Motivation and description

Motivation

The LogiKEy framework and methodology has been created to support the design, engineering, and experimentation of special-purpose reasoners, with a focus on legal-ethical reasoners, normative theories, and deontic logics. It embraces the concept of *experimentation* in formal studies where this is typically not common: by representing examples, specific domain theories, logics, and logic combinations within a computational system, we enable the formulation of predictions, their formal assessment, and the application of rigorous proof and model-finding techniques.

Given this motivation to experiment with, explore, and evaluate different theories and logics, LogiKEy provides both a unifying meta-logical framework based on semantical embeddings of object logics in expressive higher-order logic (HOL) and a systematic methodology that guides the engineering process across multiple abstraction layers. This approach allows researchers to flexibly compare alternative domain theories, assess logical variants, and iteratively refine both principles and their formal underpinnings using existing theorem provers and model finders.

An application of the LogiKEy approach is the design of ethical intelligent reasoners, where ethico-legal domain theories and deontic logics are used to enable ethical and normative reasoning with the support of computational tools based on formal methods. There, one of the main sources of motivation is the law, where the intensity of challenges has progressively increased along with the advancement of intentions to develop first legal informatics applications and then legal AI systems. This interest will just further increase in the future with the blooming of artificial intelligence: computational law for responsible AI and machine ethics aims at enabling future autonomous artificial agents to reason about normative concepts and structures in order to behave compliantly with our rules.

The generality of the LogiKEy approach is illustrated in this course by also exploring the field of computational metaphysics, which can be approached with the same methodology: automated theorem provers have been used to analyze a version of the ontological argument from Kurt Gödel's papers and revealed a previously unnoticed inconsistency in Gödel's original argument (avoided in Dana Scott's transcription).

While courses focusing on deontic logic have been given regularly at ESSLLI and have always attracted a good number of students, this is the first year where a course focused on experimentation within the LogiKEy framework and methodology is proposed. We expect that the interdisciplinary approach and novelty offered will be appealing to a wide audience.

Description: Discussed Topics

HOL and Isabelle/HOL as a Meta-Logical Workbench. We introduce classical higher-order logic (HOL) based on Church's type theory, which serves as a powerful meta-logic for representing and reasoning about other logics. Participants learn how HOL provides the expressive foundation for the LogiKEy framework and how it supports higher-order theorem proving. The Isabelle/HOL proof assistant is introduced as a practical environment for encoding definitions, writing proofs, and using automation tools such as *Sledgehammer* and *Nitpick* to perform proof search and model finding.

Shallow and Deep Embeddings of Logics in HOL. This part presents the central methodology of semantical embeddings, which allows object logics to be represented within HOL. We contrast shallow and deep embeddings, showing how propositional and multimodal logics can be formalized and automatically reasoned with inside Isabelle/HOL. Students experiment with the modal logic cube, explore frame conditions such as reflexivity and transitivity, and learn how quantified multimodal logics can be embedded.

Normative Reasoning and Deontic Logics in HOL. We explore how deontic logics—used to model obligation, permission, and prohibition—can be encoded and tested in LogiKEy. The course covers Standard Deontic Logic (SDL) and its limitations, before introducing Dyadic and Preference-based logics, such as Åqvist's System E, to handle contrary-to-duty scenarios. Rule-based approaches, including Input/Output Logic, are presented to address detachment, exceptions, and non-monotonicity. Practical examples demonstrate how these formalisms support computational reasoning about ethical and legal norms.

Computational Metaphysics: Gödel's Ontological Argument in HOL. To demonstrate the generality of the LogiKEy methodology, participants examine the formal reconstruction of Gödel's ontological argument within higher-order modal logic (HOML). Using Isabelle/HOL, we reproduce Gödel's 1970 version and Dana Scott's variant, analyzing how automated theorem provers can verify consistency, identify hidden assumptions, and reveal logical dependencies. The case study illustrates how metaphysical reasoning can be approached experimentally using the same formal tools applied in ethical and legal domains.

Logic Combinations for Agency, Knowledge, and Rights in HOL. The final part addresses how different reasoning components—such as agency, knowledge, and deontic notions—can be combined within HOL. We show how LogiKEy supports the fusion of epistemic, deontic, and action logics through modular embeddings. Participants experiment with maximal shallow embeddings to model dynamic contexts and we discuss how such combinations can represent legal and epistemic relations. As a concluding example, the course presents a case study on the dynamic logic of the Right to Know, illustrating how epistemic rights can be formalized and analyzed in LogiKEy.

3.3 Tentative outline

Monday – HOL and Isabelle/HOL. Overview. Overview of the course. The LogiKEy framework and methodology. History and interdisciplinary methodology. Introduction to classical higher-order logic (HOL) based on Church's type theory. Higher-order interactive theorem proving. Isabelle/HOL tour: types, constants, definitions, abbreviations, functions, proof automation with *Sledgehammer* and (counter-)model finding with *Nitpick*, when to automate vs. interact. HOL as a meta-logical language.

Tuesday – Shallow and Deep Embeddings in HOL. Shallow Semantical Embeddings (SSE) of object logics in the meta-logic HOL. Deep vs. shallow embeddings of logics. Propositional logic embedding demo. The modal logic cube in Isabelle/HOL.

Frame conditions and testing axioms in Isabelle/HOL. Propositional multimodal logics via SSE. First-order multimodal logics via SSE. Automation of faithfulness proofs; converting formulas across embeddings.

Wednesday – Normative Reasoning: Deontic Logics in HOL. Philosophical challenges and foundations of Normative Reasoning. Deontic paradoxes. Embedding of Standard Deontic Logic as the modal logic KD, its semantics, and main questions. Embedding of Preference and Dyadic Deontic Logics to address the problem of Contrary-to-Duty. Embedding of Input/Output Logic to address the problem of Detachment. Explaining normative decisions. Reasoning with values.

Thursday – Experiments in Metaphysics: Gödel's Ontological Argument in HOL. Higher-order multimodal logic (HOML). SSE of HOML in HOL. Gödel's modal ontological argument as presented in his 1970 manuscript. Isabelle/HOL encoding of Gödel's ontological argument. Gödel's notion of positive properties, and how it constitutes a (modal) ultrafilter. Scott's variant of the argument, and Isabelle/HOL encoding.

Friday – Logic Combinations in HOL to Reason about Agency, Knowledge, and More. Wrapping up. LogiKEy and other topics in Knowledge Representation: encoding agents, actions, knowledge, and practical reasoning. Legal relations and the problem of formalizing rights. Combining logics in the meta-logic HOL. Epistemic-deontic combinations. Maximal shallow embeddings to capture evaluation domains that can vary in the evaluation of a formula. Case study: embedding of a dynamic logic of the Right to Know in HOL. Comments on open problems, unification, wrapping up and outlook.

3.4 Expected level and prerequisites

In order for the participants to benefit from this course, we expect them to have a basic understanding of discrete mathematics, propositional logic and modal logic.

3.5 Selected references

- Christoph Benzmüller, Xavier Parent, Leendert van der Torre: Designing normative theories for ethical and legal reasoning: LogiKEy framework, methodology, and tool support. *Artificial Intelligence 287* (2020).
(`https://doi.org/10.1016/j.artint.2020.103348`)

- Christoph Benzmüller, Ali Farjami, David Fuenmayor, Paul Meder, Xavier Parent, Alexander Steen, Leendert van der Torre, Valeria Zahoransky: *LogiKEy workbench: Deontic logics, logic combinations and expressive ethical and legal reasoning (Isabelle/HOL dataset).* Data in Brief, Volume 33, 2020.
(`https://doi.org/10.1016/j.dib.2020.106409`)

- Christoph Benzmüller, Peter Andrews: *Church's Type Theory.* In Zalta, E. N., and Nodelman, U. (eds.), *The Stanford Encyclopedia of Philosophy.* Metaphysics Research Lab, Stanford University, Spring 2024 edition, 2024.
(`https://plato.stanford.edu/archives/spr2024/entries/type-theory-church/`)

- Christoph Benzmüller: *Faithful Logic Embeddings in HOL — Deep and Shallow.* In Barrett, C., and Waldmann, U. (eds.), *Automated Deduction – CADE-30 –*

30th International Conference on Automated Deduction, Stuttgart, Germany, July 28–31, 2025, Proceedings, volume 15943 of Lecture Notes in Computer Science, pages 280–302, 2025. Springer.
(preprint: `https://arxiv.org/abs/2502.19311`)

- Christoph Benzmüller, Marvin Claus, Naiba Sultana: *Systematic Verification of the Modal Logic Cube in Isabelle/HOL.* In Kaliszyk, C., and Paskevich, A. (eds.), *PxTP 2015*, volume 186, pages 27–41, Berlin, Germany, 2015. *EPTCS.*
(`https://doi.org/10.4204/EPTCS.186.5`)

- Xavier Parent and Leon van der Torre: *Introduction to Deontic Logic and Normative Systems.* College Publications, 2018.
(`http://www.collegepublications.co.uk/TLR/?00001`)

- Dov Gabbay, John Horty, Xavier Parent, Ron van der Meyden, and Leendert van der Torre (eds.): *Handbook of Deontic Logic and Normative Systems*, volume 1. College Publications, 2013.
(`http://www.collegepublications.co.uk/handbooks/?00001`)

- Dov Gabbay, John Horty, Xavier Parent, Ron van der Meyden, and Leendert van der Torre (eds.): *Handbook of Deontic Logic and Normative Systems*, volume 2. College Publications, 2021.
(`http://www.collegepublications.co.uk/handbooks/?00005`)

- Christoph Benzmüller, Ali Farjami, and Xavier Parent: *Åqvist's Dyadic Deontic Logic E in HOL. Journal of Applied Logics – IfCoLoG Journal of Logics and their Applications* (Special Issue: Reasoning for Legal AI), 6(5):733–755, 2019.
(`https://www.collegepublications.co.uk/ifcolog/?00034`)

- Christoph Benzmüller, David Fuenmayor, and Bernhard Lomfeld: *Modelling Value-oriented Legal Reasoning in LogiKEy. Logics*, 2(1):31–78, 2024.
(`https://doi.org/10.3390/logics2010003`)

- Christoph Benzmüller, Dana Scott: *Notes on Gödel's and Scott's Variants of the Ontological Argument. Monatshefte für Mathematik*, 2025.
(`https://doi.org/10.1007/s00605-025-02078-x`)

- Christoph Benzmüller, Bruno Woltzenlogel Paleo: *Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers.* In Schaub, T., Friedrich, G., and O'Sullivan, B. (eds.), *ECAI 2014*, Frontiers in Artificial Intelligence and Applications, volume 263, IOS Press, Amsterdam, 2014, pages 93–98.
(`https://doi.org/10.3233/978-1-61499-419-0-93`)

- Lara Lawniczak, Luca Pasetto, Christoph Benzmüller, Xu Li, Réka Markovich: *Reasoning with Epistemic Rights and Duties: Automating a Dynamic Logic of the Right to Know in LogiKEy. 28th European Conference on Artificial Intelligence (ECAI 2025)*, (accepted/to appear).

4 Information required of course proposers

- Besides students interested in logic and computer science, the course will appeal to students outside of its main discipline because the topic is highly inter-disciplinary and it links with other fields such as machine ethics, legal theory, AI&Law, and philosophy.

- Christoph Benzmüller has 30 years of experience in academic teaching at reknown universities in Germany and abroad. His courses cover topics in Logic, Automated Reasoning, Computer Science, Artificial Intelligence and Computational Philosophy. He is also experienced as an invited speaker with more than 100 invited presentations over the past two decades. Benzmüller, supported by colleagues and students, has held tutorials and seminars on LogiKEy and Computational Metaphysics in five continents.

- Luca Pasetto has experience in teaching foundational subjects such as *Discrete Mathematics 1*, a first-year bachelor's course at the Department of Computer Science of the University of Luxembourg, and more advanced courses such as *Selected topics in Artificial Intelligence* at the master's level. Luca has also experience in an interdisciplinary setting: at the University of Verona, he has taught courses of computer science for non computer scientists at the Department of Foreign Languages and Literatures, and he has assisted courses such as Logic, Programming 1, and Signal and Image Processing. He also has experience in presenting an ESSLLI course, as he taught the course *Deontic Logic, Normative Systems, and Their Application in AI&Law* at ESSLLI 2025.

5 Other information

Luca Pasetto expects to be able to finance his own expenses.